



Lab 2: Hands-On Git and Version Control

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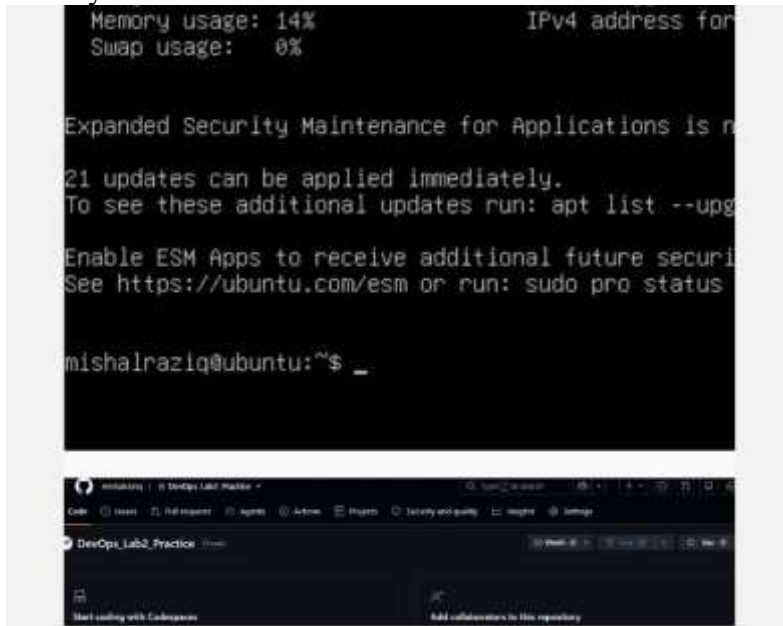
Submitted to : Sir Waqas

Department of Computer Science

Task 1 — Create Private GitHub Repository

1. repo_private.png

Caption: Private DevOps_Lab2_Practice repository created successfully on GitHub with the required terminal identity visible.



Task 2 — Connect Repository via SSH

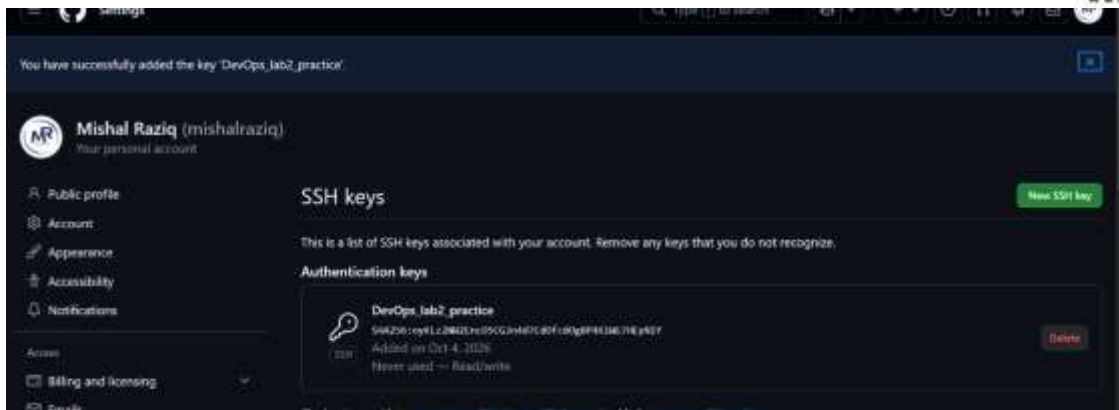
2. ssh_keygen.png

Caption: SSH key generated successfully and the public key displayed without exposing the private key.



3. github_sshkey.png

Caption: SSH public key successfully added to the GitHub account.



4. ssh_clone.png

Caption: DevOps Lab2 Practice repository successfully cloned using SSH.

```
mishalraziq@ubuntu:~$ git clone git@github.com:mishalraziq/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbp2IsF/zLDA0zPMSvHdkr4UvCOQu
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
warning: You appear to have cloned an empty repository.

mishalraziq@ubuntu:~$
mishalraziq@ubuntu:~$ cd DevOps_Lab2_Practice
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git remote -v
origin  git@github.com:mishalraziq/DevOps_Lab2_Practice.git (fetch)
origin  git@github.com:mishalraziq/DevOps_Lab2_Practice.git (push)
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 3 — Configure Git Username and Email

5. git_identity.png

Caption: Git username and email successfully configured for the local repository environment.

```
origin  git@github.com:mishalraziq/DevOps_Lab2_Practice.git (fetch)
origin  git@github.com:mishalraziq/DevOps_Lab2_Practice.git (push)
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.name "mishalraziq"
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.email "mishalraziq789@gmail.com"
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

6. git_config_list.png

Caption: Git configuration verified successfully using git config --list.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git config --list
user.name=mishalraziq
user.email=mishalraziq789@gmail.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
remote.origin.url=git@github.com:mishalraziq/DevOps_Lab2_Practice.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 4 — Explore .git Folder

7. git_folder.png

Caption: Hidden .git directory and its internal Git files successfully displayed.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ ls -a
.  ..  .git
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ ls -a .git
.  ..  HEAD  config  description  hooks  info  objects  refs
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 5 — Local Repository Management

8. delete_git.png

Caption: Existing .git directory removed and Git reports that the directory is no longer a repository.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ rm -rf .git
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git status
fatal: not a git repository (or any of the parent directories): .git
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

9. git_init.png

Caption: Local repository successfully re-initialized using git init.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/mishalraziq/DevOps_Lab2_Practice/.git/
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

10. first_commit.png

Caption: README.md successfully added and committed as the initial commit.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ echo "#Lab02 Git Practice" >> README.md
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git add README.md
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Initial Commit"
[master (root-commit) fb244bd] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

11. first_push.png

Caption: Local main branch successfully connected to and pushed to GitHub.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git remote add origin git@github.com:mishalraziq/DevOps_Lab2_Practi
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git branch -M main
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 235 bytes | 19.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 6 — File Status & Staging

12. status1.png

Caption: Git status shows the newly created notes.txt file as an untracked file.'

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ echo "Adding some text notes" >> notes.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    notes.txt

nothing added to commit but untracked files present (use "git add" to track)
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

13. commit_notes.png

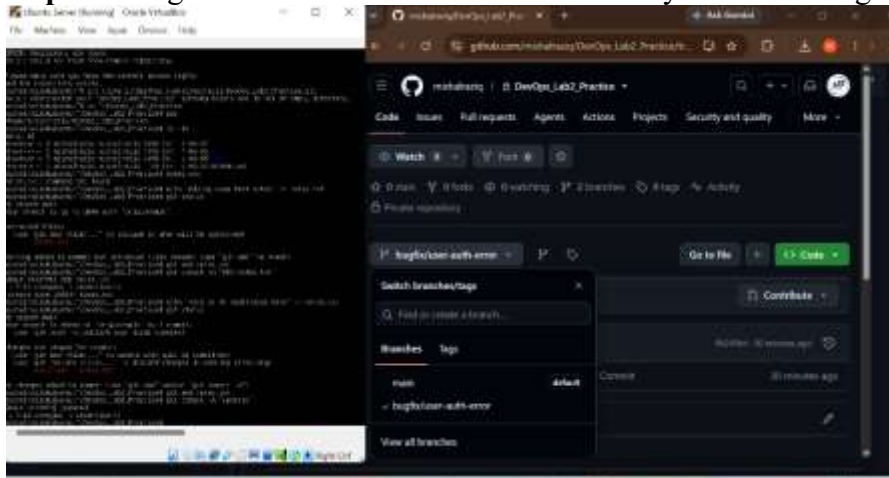
Caption: notes.txt successfully staged and committed to the repository.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git add notes.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "ADD notes.txt"
[main 3f4a791] ADD notes.txt
1 file changed, 1 insertion(+)
create mode 100644 notes.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 7 — Branch Creation Using GitHub GUI

14. bugfix_branch_gui.png

Caption: bugfix/user-auth-error branch successfully created through the GitHub web interface.



15. bugfix_branch_local.png

Caption: The GitHub-created bugfix branch successfully synchronized and displayed in the local branch list.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git pull origin bugfix/user-auth-error
From github.com:mishalraziq/DevOps_Lab2_Practice
* branch          bugfix/user-auth-error -> FETCH_HEAD
* [new branch]     bugfix/user-auth-error -> origin/bugfix/user-auth-error
Already up to date.
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git branch
* main
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git branch
* main
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 8 — Branch Creation and Push Using Git Bash

16. feature_db_branch.png

Caption: feature/db-connection branch successfully created locally and pushed to GitHub.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ cat db_connection.txt
database feature added for lab2
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git add db_connection.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature/db-connection
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   db_connection.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "add feature"
[feature/db-connection a1dd4ef] add feature
1 file changed, 1 insertion(+)
create mode 100644 db_connection.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 9 — Branching & Merging

17. branch_create.png

Caption: feature-1 branch successfully created and checked out from the main branch.

```
mishalraziq@ubuntu:~$ cd ~/DevOps_Lab2_Practice
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ _
```

18. feature_commit.png

Caption: New functionality successfully added to main.py and committed on the feature branch.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ cat main.py
def hello():
    print("hello from lab2")
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature-1
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   main.py
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "ADD NEW FUNCTION TO main.py"
[feature-1 bfd84d3] ADD NEW FUNCTION TO main.py
 1 file changed, 2 insertions(+)
 create mode 100644 main.py
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

19. merge.png

Caption: feature-1 successfully merged into the main branch.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Updating ce790f8..bfd84d3
Fast-forward
 main.py | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 main.py
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

20. push_branches.png

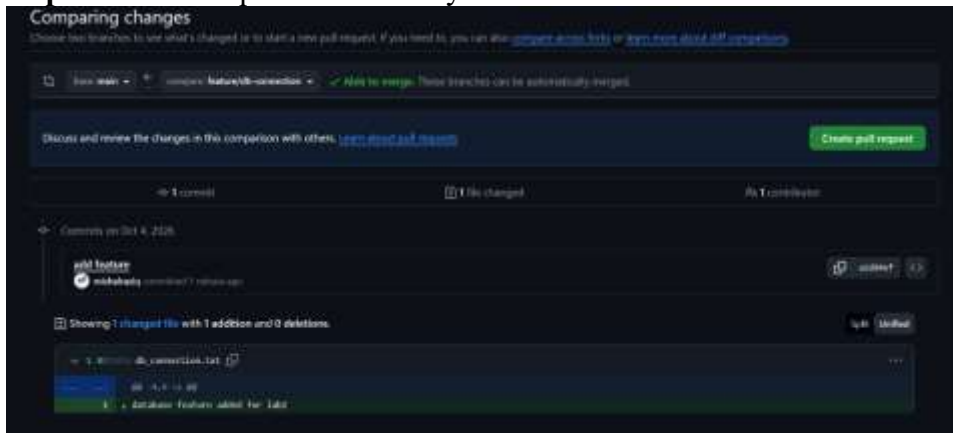
Caption: Updated main and feature-1 branches successfully pushed to the remote repository.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 366 bytes | 33.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 fb244bd..bfd84d3  main -> main
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin feature-1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-1' on GitHub by visiting:
remote:   https://github.com/mishalraziq/DevOps_Lab2_Practice/pull/new/feature-1
remote:
To github.com:mishalraziq/DevOps_Lab2_Practice.git
 * [new branch]   feature-1 -> feature-1
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 10 — Pull Request and Branch Review

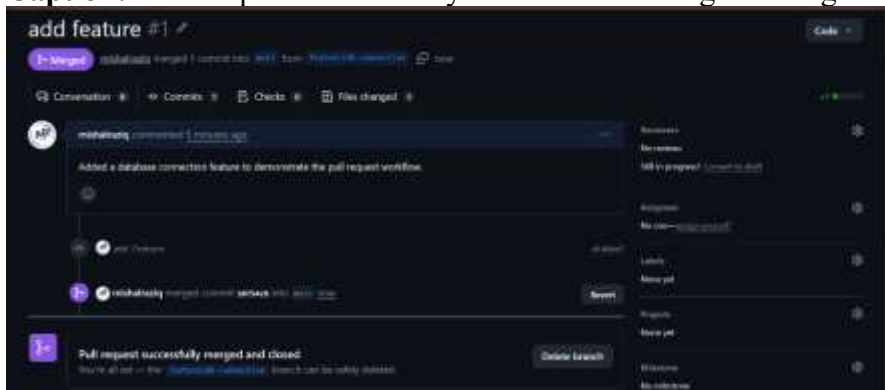
21. pr_creation.png

Caption: Pull Request successfully created from feature/db-connection into main.



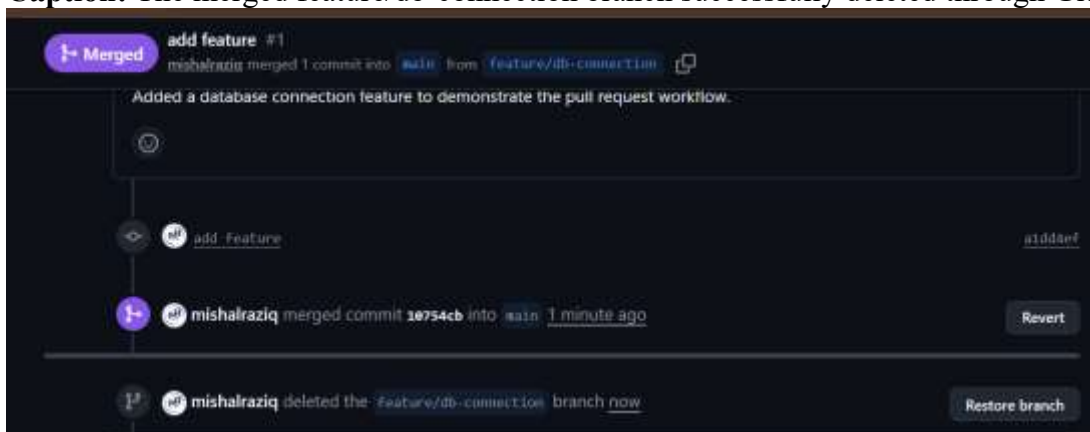
22. pr_merge.png

Caption: Pull Request successfully reviewed and merged through the GitHub interface.



23. branch_delete.png

Caption: The merged feature/db-connection branch successfully deleted through GitHub.



Task 11 — Detailed Branch Strategy

24. branch_strategy.png

Caption: Professional branching structure with main, develop, staging, and feature/bugfix branches successfully established.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git branch
develop
feature-1
* main
staging
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

25. branch_merges.png

Caption: Feature work successfully merged into develop and subsequently into staging.

```
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git add strategy.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m'added'
[develop f9a0f86] added
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin develop
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 562 bytes | 40.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:mishalraziq/DevOps_Lab2_Practice.git
  10754cb..f9a0f86 develop -> develop
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout staging
Switched to branch 'staging'
Your branch is up to date with 'origin/staging'.
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git merge develop
Updating 10754cb..f9a0f86
Fast-forward
 strategy.txt | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 strategy.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin staging
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mishalraziq/DevOps_Lab2_Practice.git
  10754cb..f9a0f86 staging -> staging
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

26. final_merge.png

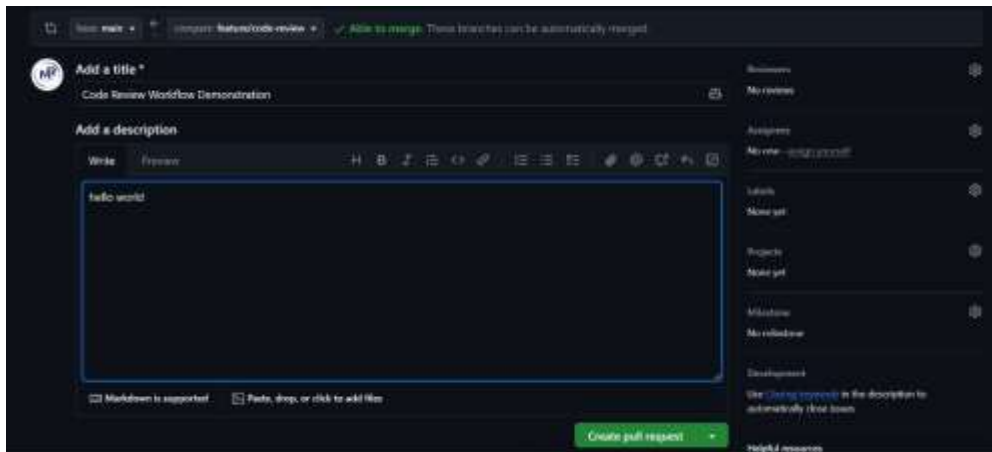
Caption: Tested changes successfully promoted from staging into the main production branch.

```
10754cb..f9a0f86 staging -> staging
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git merge staging
Updating 10754cb..f9a0f86
Fast-forward
 strategy.txt | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 strategy.txt
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin main
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:mishalraziq/DevOps_Lab2_Practice.git
  10754cb..f9a0f86 main -> main
mishalraziq@ubuntu:~/DevOps_Lab2_Practice$
```

Task 12 — Code Review Workflow

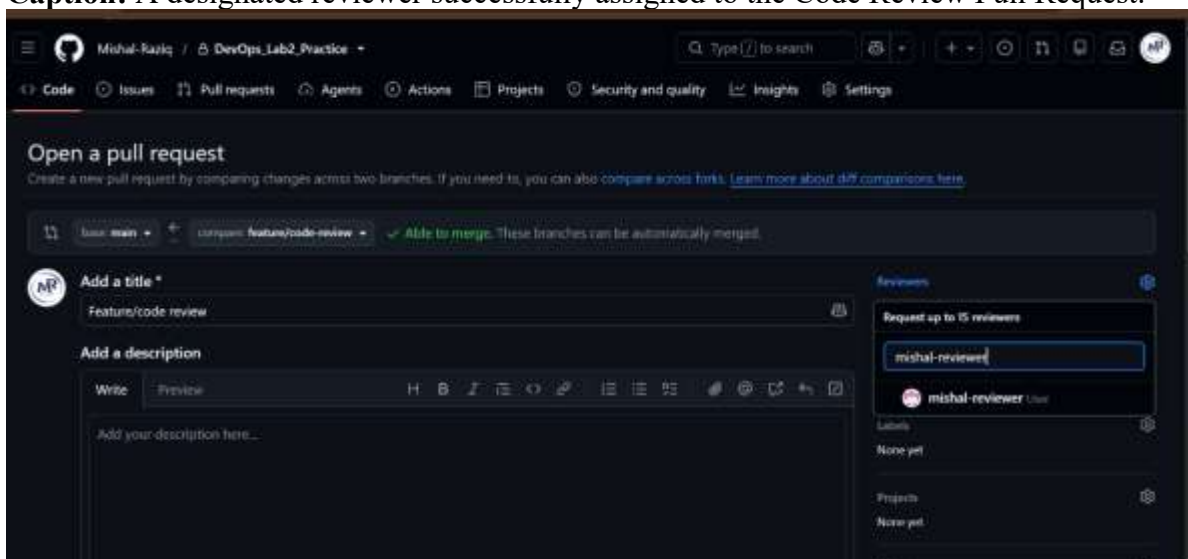
27. pr_create_details.png

Caption: Code Review Pull Request created with a clear title, description, and feature-to-main branch comparison.



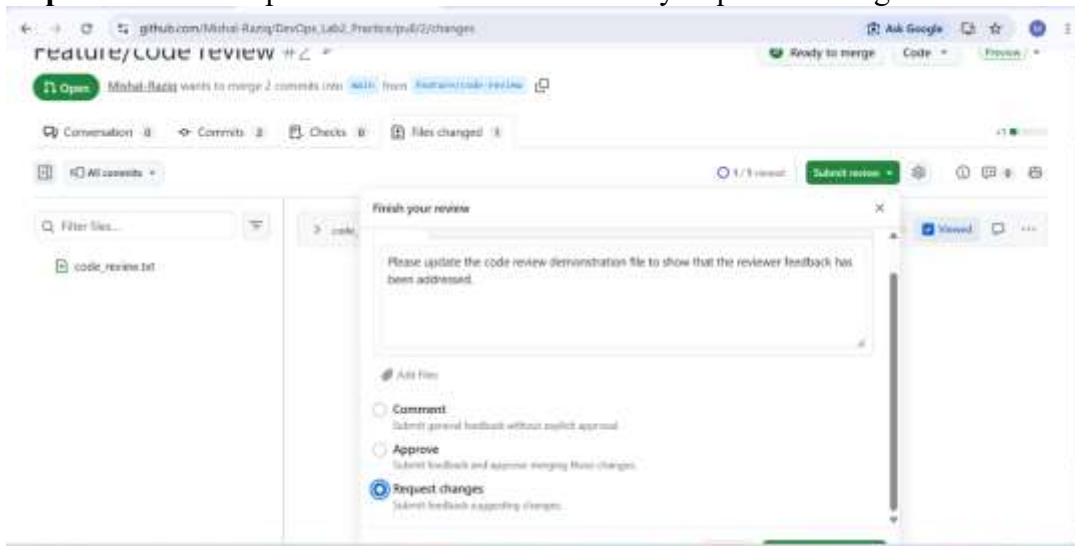
28. pr_assigned_reviewer.png

Caption: A designated reviewer successfully assigned to the Code Review Pull Request.



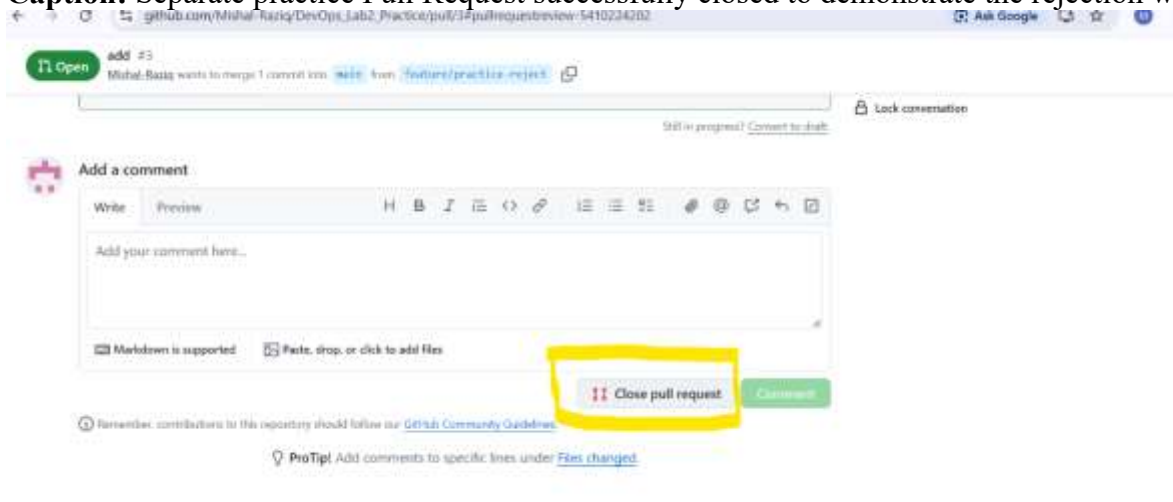
29. pr_request_changes.png

Caption: Reviewer provided feedback and formally requested changes on the Pull Request.



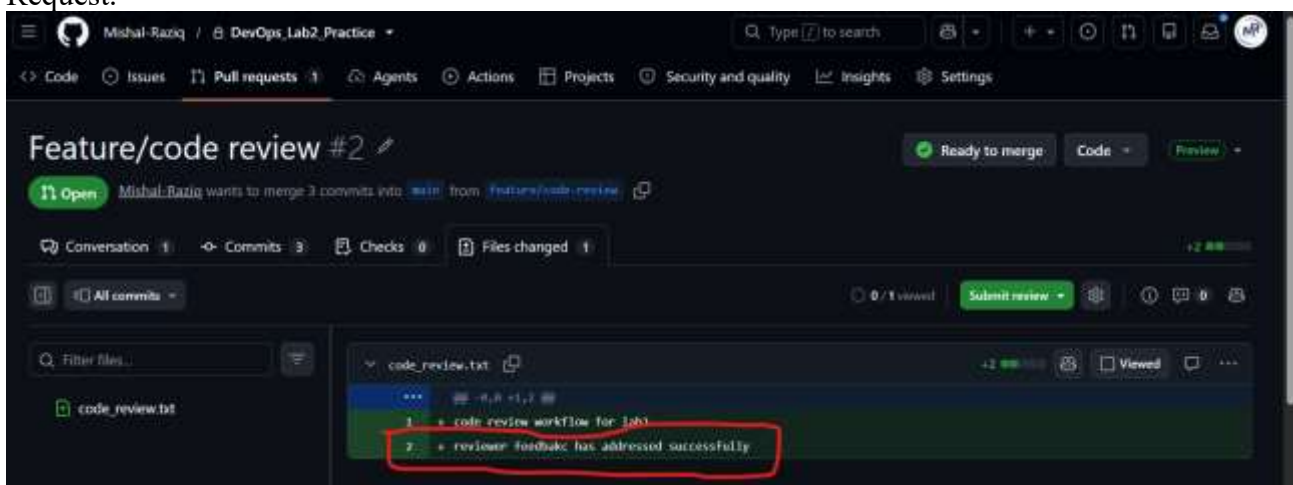
30. pr_rejected.png

Caption: Separate practice Pull Request successfully closed to demonstrate the rejection workflow.



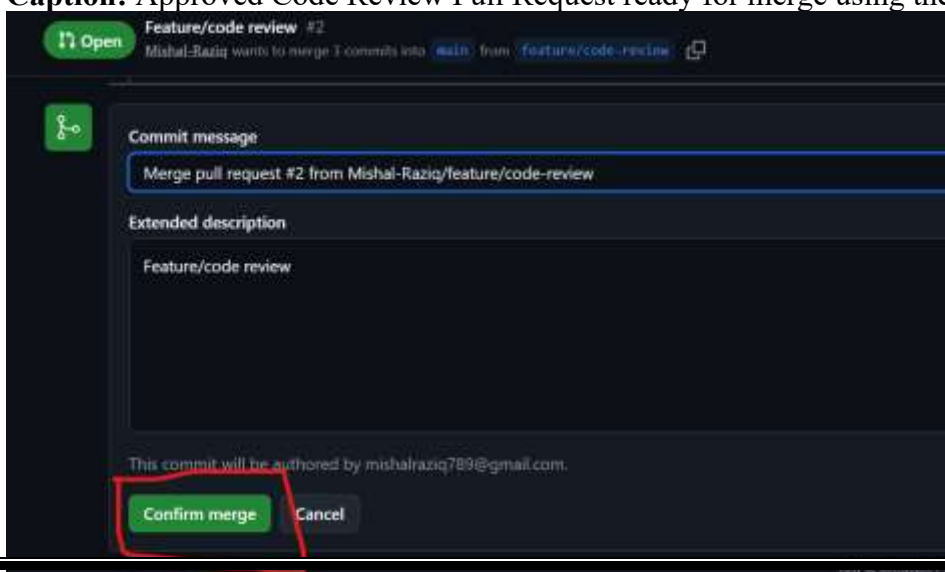
31. pr_updated_with_commits.png

Caption: Author addressed reviewer feedback and pushed new commits, automatically updating the Pull Request.



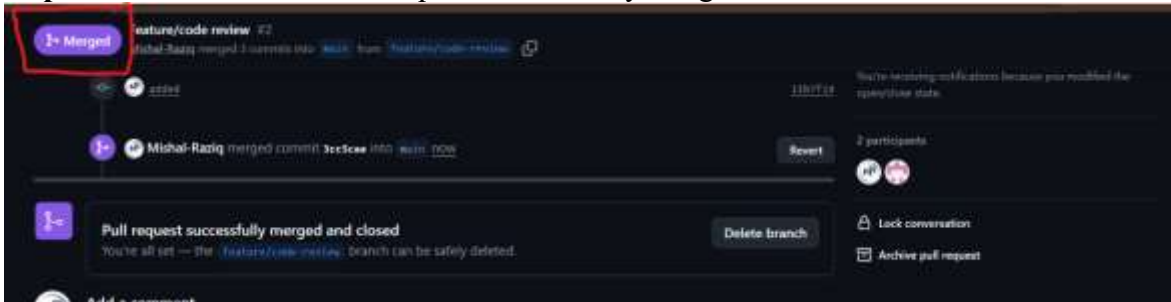
32. pr_merge_confirm.png

Caption: Approved Code Review Pull Request ready for merge using the selected merge strategy.



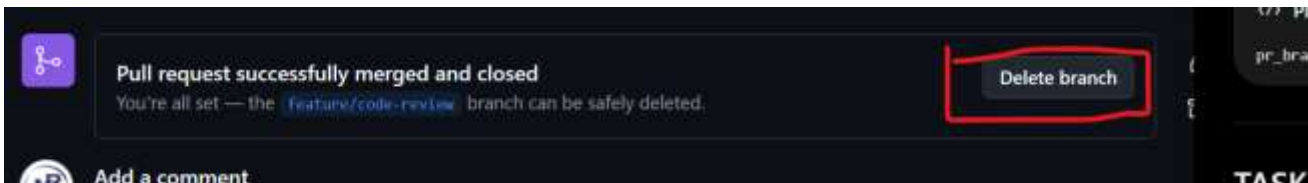
33. pr_merged.png

Caption: Code Review Pull Request successfully merged into the main branch.



34. pr_branch_deleted.png

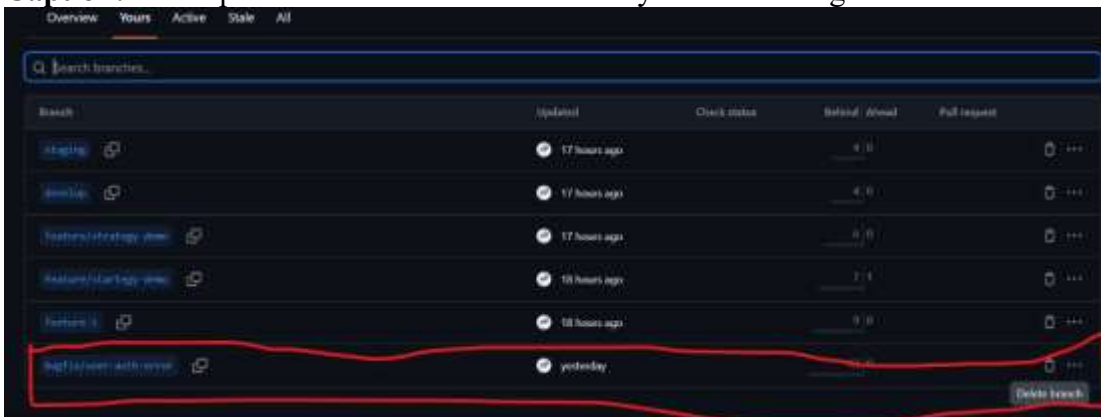
Caption: The merged Code Review feature branch successfully deleted after completion.



Task 13 — Branch Cleanup Best Practices

35. remote_branch_deleted.png

Caption: A completed remote branch successfully deleted through the GitHub branch interface.



36. remote_branch_delete_cmd.png

Caption: A completed remote branch successfully deleted from the terminal using git push origin --delete.

```

mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete feature/practice-reject
remote: This repository moved. Please use the new location:
remote:   git@github.com:Mishal-Raziq/DevOps_Lab2_Practice.git
To github.com:mishalraziq/DevOps_Lab2_Practice.git
- [deleted]          feature/practice-reject
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
  
```

Q1 —

Q1_branch_created.png —

Create Q1 Branch: Q1 feature branch feature/q1-demo created successfully.

```
mishal-raziq@ubuntu: ~/DevOps_Lab2_Practice$
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature/q1-demo
Switched to a new branch 'feature/q1-demo'
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git branch
  develop
  feature-1
  feature/code-review
  feature/practice-reject
* feature/q1-demo
  feature/strategy-demo
  main
  staging
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

Q1_commit_done.png —

Commit Q1 Changes: Q1 demonstration file created and committed successfully.

```
staging
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ echo "Q1 MERGE " > q1_demo.txt
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ cat q1_demo.txt
Q1 MERGE
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git add q1_demo.txt
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git commit -m"added"
[feature/q1-demo 6d1868d] added
 1 file changed, 1 insertion(+)
 create mode 100644 q1_demo.txt
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

Q1_merge_done.png —

Merge Q1 Branch: feature/q1-demo successfully merged into main.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git merge feature/q1-demo
Updating 3cc5caa..6d1868d
Fast-forward
 q1_demo.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 q1_demo.txt
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```

Q1_merge_history.png —

Verify Merge History: Git log shows the Q1 commit and merge history successfully.

```
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph --all
* 6d1868d (HEAD -> main, feature/q1-demo) added
*   3cc5caa (origin/main, origin/HEAD) Merge pull request #2 from Mishal-Raziq/feature/code-rev:
  \
  * 11b7f14 (origin/feature/code-review, feature/code-review) added
  * 6f682c8 added
  * 04b3211 added
  /
*   4f59a99 (feature/practice-reject) add
  \
  * f9a0f86 (origin/staging, origin/develop, staging, develop) added
  /
*   4d16375 (origin/feature/strategy-demo, feature/strategy-demo) updated
* | eacbc43 add
  /
*   0e06e3d (origin/feature/strategy-demo) add branching feature
  /
*   10754cb Merge pull request #1 from mishalraziq/feature/db-connection
  \
  * a1dd4ef (origin/feature/db-connection) add feature
  * | bfd04d3 (origin/feature-1, feature-1) ADD NEW FUNCTION TO main.py
  /
* ce790f8 updated
* 3f4a791 ADD notes.txt
* fb244bd (origin/bugfix/user-auth-error) Initial Commit
mishal-raziq@ubuntu:~/DevOps_Lab2_Practice$
```